

Please provide a 1,250-3,250 character statement discussing your educational path (approx. 250 - 650 words). How does continuing your education at a new institution help you achieve your future goals?

I did not come into college with a complete plan. I came in with a curiosity about AI that had already pulled me years ahead of where I was supposed to be.

Before I had any formal instruction, I was building. I engineered a real-time conversational AI pipeline achieving 1.2 second response times on consumer hardware, published models that have been downloaded over 2,600 times on HuggingFace, and competed in Google hackathons while still in high school. None of that came from a classroom. It came from my obsessive need to understand how things worked and a willingness to figure it out entirely on my own.

College was supposed to be where I formalized that curiosity. And in some ways it has been. I have taken on a technical practicum leading AI development for the Northeastern University and Mills College Art Museum, I am authoring a research paper on a novel inference method I developed called Post-Thinking, and I have been consulting local businesses in Oakland on AI integration through Northeastern's AI clinic. I have proven I can do the work. What I have not found is an environment that meets me where I am and pushes me further.

The gap in my education is not on the software side but rather it is on the hardware side. I have spent years building on top of systems I do not fully understand at the physical level. I know how to optimize inference pipelines. I do not know what is happening inside the chip when I do it. Closing that gap is not just an academic interest for me. It is the difference between being someone who builds AI systems and someone who can engineer the full stack they run on, which is exactly the kind of engineer I intend to become.

Cornell ECE is a place where that is possible. The program sits precisely at the intersection of hardware and software in a way that matches how I already think about problems. I would not be starting over. I simply seek to add the foundation that makes everything I have already built more complete. My goal is to work on the systems that make AI faster, more efficient, and more deployable in the real world. Cornell is where I get the tools to do that.